DSE 316 Project Proposal
SAE-Steer: SAE-based Steering of Fine-Grained Visual Concepts in Latent Diffusion Models

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1 Abstract

While Latent Diffusion Models (LDMs) excel at high-fidelity image synthesis, achieving granular, mechanistic control over specific visual attributes remains a challenge due to the polysemantic nature of the model's internal representations. Recent advances in Mechanistic Interpretability have introduced Sparse Autoencoders (SAEs) as a tool to disentangle these representations into monosemantic features. We propose a framework to steer fine-grained visual concepts such as specific animal breeds or localized textures by performing causal interventions in the SAE latent space [1]. By leveraging layer-wise insights and feature attribution methods, this project aims to enable precise, zero-shot attribute editing without the need for auxiliary models or costly fine-tuning.

2 Problem Statement

Current diffusion editing techniques (ex : prompt engineering, ControlNet, or LoRA) operate as "black boxes." When a user attempts to modify a specific attribute, the model often suffers from **attribute leakage** or **concept bleeding**, where changing one feature (ex : "adding a beard") unintentionally alters unrelated features (ex : "changing the background" or "altering age"). This occurs because neurons in the U-Net are polysemantic, i.e, they represent multiple unrelated concepts simultaneously. The problem is to identify and isolate individual, monosemantic "switches" within the model's latent space to allow for orthogonal control over visual concepts.

3 Motivation

"To understand a system, one must be able to intervene in its internal state and predict the outcome." While existing literature has used SAEs for observation (*Revelio*) or total concept removal (*SAeUron*), there is a lack of frameworks for **compositional steering**.

The novelty of this project lies in the **Mechanistic Intervention Pipeline**. Unlike previous works that focus on global biases, we propose to target the `up_ft1` and bottleneck layers to perform "surgical" style swaps. By identifying the causal features for fine-grained concepts and scaling them in the latent space during the denoising process, we enable a level of precision control that exceeds standard text-prompting.

4 Literature Review

This project builds upon three foundational papers:

- **Revelio [1]:** Proved that k-SAEs can uncover monosemantic features in diffusion models and mapped specific semantic granularities to different U-Net layers.[1]
- **DiffLens [2]:** Demonstrated that scaling SAE features can mitigate social biases (gender, race) in a zero-shot manner.[2]
- **SAeUron [3]:** Introduced score-based feature selection to identify and ablate (zero-out) unwanted styles or concepts.[3]

5 Proposed Methodology & Algorithms

The framework involves three technical phases:

1. **Feature Discovery:** We aim to utilize k-SAEs to project U-Net activations x into a sparse latent space f:

$$f = \text{ReLU}(W_{enc}(x - b_{pre}) + b_{enc}) \tag{1}$$

where the top-k activations are retained to ensure monosemanticity.

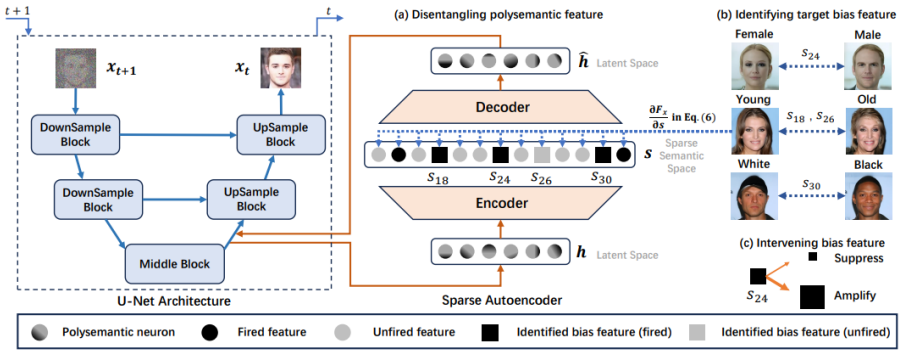

Figure 1: A Schematic showing how k-SAEs disentangle polysemantic neurons. Image from [2]

2. **Causal Identification:** We will use **Gradient-based Attribution** [2] to find the index i of the feature that maximizes the response for a target concept C:

$$i = \text{argmax}_j \left(\frac{\partial \mathcal{L}_{prompt}}{\partial f_j} \right) \tag{2}$$

3. **Inference-Time Steering:** During the reverse diffusion process ($T \in [200, 600]$), we apply a steering vector to the SAE latents:

$$f'_i = f_i + \alpha, \quad x' = W_{dec} f' + b_{dec} \tag{3}$$

The modified activation x' is re-inserted via a forward hook.

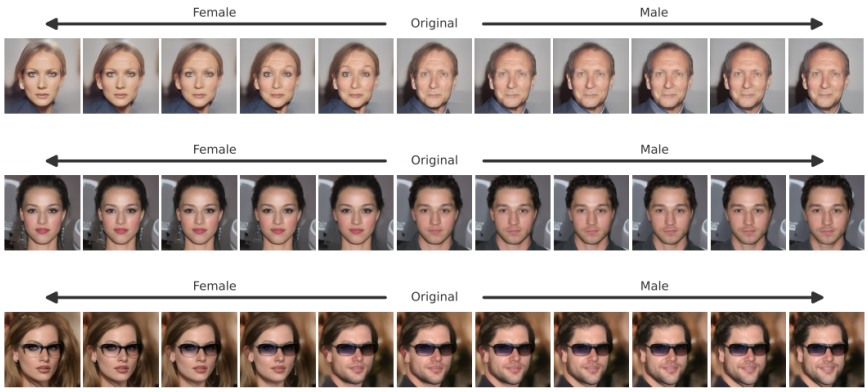

Figure 2: A sample of inference time steering [2]

6 Expected Output & Evaluation

We expect to produce a "steering dashboard" where individual SAE features can be toggled to change image attributes (ex : changing a "Siamese cat" to a "Persian cat") while keeping the pose and background constant.

- **Metrics:** We will use **CLIP-Score** for semantic alignment and **LPIPS** for structural preservation.
- **Visualization:** Feature activation maps will be generated to prove the locality of the intervention.

7 Relevance to DSE 316 Deep Learning

This project applies core course concepts in a state-of-the-art context:

- **Latent Representation Learning:** Manipulating high-dimensional manifold projections.
- **Sparsity and Regularization:** Utilizing the k-sparsity constraint of SAEs to achieve monosemanticity.
- **Backpropagation Mechanics:** Using gradients for attribution and steering.

8 Team Members and Roles

- **J Harishwar Rao:** Project Lead. Responsible for SAE implementation, hook injection in the Diffusers library.
- **Komal Sandeep:** Data Engineering. Responsible for sourcing fine-grained datasets and compute pipeline management.
- **Uddisa Kkaarthiek:** Evaluation & Metrics. Responsible for CLIP/LPIPS scores and qualitative monosemanticity checks.

References

[1] Kim et al. "Revelio: Interpreting and leveraging semantic information in diffusion models." *arXiv:2411.16725*.

[2] Shi et al. "Dissecting and Mitigating Diffusion Bias via Mechanistic Interpretability." *arXiv:2503.20483*.

[3] Cywiński and Deja. "SAeUron: Interpretable Concept Unlearning in Diffusion Models with Sparse Autoencoders." *arXiv:2501.18052*.